

Rudresh Singh

Backend Software Developer

Varanasi, India • +91 8467931976 • rslikefoot00@gmail.com • [LinkedIn](#) • [GitHub](#)

SUMMARY

I'm a backend engineer who thinks like an architect — I plan, sometimes overthink, but when I lock in, I ship. I'm drawn to distributed systems, infrastructure, and AI orchestration. Not because it's trendy these days, but because I genuinely love designing things, orchestrating that work at scale. Looking for a team where I can go deep, learn from people who've been there, and grow into the engineer I'm working toward.

PROJECTS

BodhAPI — Indian News Intelligence Engine — *Node.js · Express 5 · PostgreSQL · Redis · Docker · AWS EC2 · Nginx · Gemini API* [↗ GitHub](#) [↗ Live](#)

- Built a production REST API with 16 endpoints serving structured Indian news — filtering by category, sentiment, state, entity, emergency type, and full-text search.
- Reduced endpoint latency **from ~300ms to under 20ms** via Redis TTL caching; designed Lua-based atomic token bucket rate limiter with per-minute and daily quotas.
- Implemented dual authentication: **JWT (access + refresh token rotation)** and **API key auth** using SHA-256 prefixed tokens stored as bcrypt hashes — keys safe even if DB is compromised. Prefix indexed for fast lookup.
- PostgreSQL performance layer: BRIN, GIN, B-Tree, composite indexes + monthly table partitioning. IP-based rate limiting offloaded to Nginx.
- Orchestrated **4 Docker containers (API, scraper, PostgreSQL, Redis) under 1GB RAM** on a swap-configured EC2 instance — private Docker network isolates internal services; only Nginx is exposed.
- Managed schema evolution with **dbmate migrations** + pg_dump baseline; automated Docker image lifecycle with custom push.sh/pull.sh scripts via Docker Hub.

Bodh — AI News Ingestion Pipeline — *Node.js · Playwright · BullMQ · YouTube API · Gemini API · PostgreSQL* [↗ GitHub](#)

- Built a **multi-stage AI ingestion pipeline**: scheduler → YT API channel fetch (filtered by regex, duration, freshness) → Playwright transcript scraper → Gemini AI structuring → PostgreSQL.
- Implemented fault-tolerant worker architecture: transcripts saved to DB with **is_used flag** before AI processing — prevents data loss on failure. Separate workers for scraping, AI structuring, and final DB save.
- Gemini API processes transcripts into structured JSON (category, sentiment, entities, emergency type, financial data, importance score) which is then bulk-inserted into partitioned PostgreSQL tables.

Saraansh — AI YouTube Summarizer (Chrome Extension) — *JavaScript · Chrome Extension APIs · AI API* [↗ GitHub](#) [↗ Live](#)

- Developed a Chrome extension that extracts YouTube transcripts and generates AI summaries, now **used by 100+ students** at college.
- Applied SOLID principles throughout the modular architecture; built multiple iterations improving stability and transcript extraction logic.

Niriksana — OSINT Geospatial Platform (WIP) — *React · TypeScript · GeoJSON* [↗ GitHub](#)

- Building an interactive geospatial intelligence platform with Indian administrative boundary visualization. Backend (Node.js, spatial querying, auth) in active development.

TCP Chat Server — *C++ · Winsock* [↗ GitHub](#)

- Implemented a multi-client TCP chat server from scratch in C++ using Winsock — covering socket programming, concurrent client handling, and server architecture fundamentals.

TECHNICAL SKILLS

Backend Node.js, Express 5, REST API design, middleware architecture, rate limiting, background job processing

Databases PostgreSQL (partitioning, GIN/BRIN/B-Tree indexes, full-text search, JSONB, transactions), Redis (caching, TTL, Lua scripting, queues), MongoDB

Auth & Security JWT (access + refresh token rotation), Google OAuth 2.0, Passport.js, bcrypt hashing, SHA-256 prefixed API keys, input sanitization, parameterized queries

DevOps & Cloud Docker, Docker Compose, Nginx (reverse proxy + IP rate limiting), AWS EC2, HTTPS/SSL, dbmate migrations, Docker Hub, Linux (swap config)

Languages JavaScript (ES6+), TypeScript, C++, SQL, Bash

Frontend React.js, React Native (basics), CORS, GitHub Pages

Tools Playwright, BullMQ, YouTube Data API, Gemini API, Git, pg_dump

EDUCATION

B.Tech — Computer Science (AI/ML specialization) | Kashi Institute of Technology, AKTU | Sep 2023 – Oct 2027

ACHIEVEMENTS

-
- HackerRank 5★ C++ — built strong fundamentals in OOP, STL, data structures, exception handling.
 - 200+ DSA problems solved across different platforms (arrays, hashmaps, stacks, trees) — actively solving.
 - Saraansh extension adopted organically by 100+ college students.